Comparing Mutual Information in Finite Samples

Can MI-specific degrees of freedom improve comparisons between populations?

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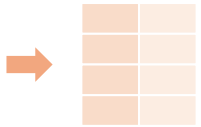
Motivation: comparing MI across populations

Mutual information measures how much observing one variable reduces uncertainty about another, on average.
Different sample estimates can arise from sampling noise.

Hypotheses $H_0 : I(P) = I(Q)$ against $H_1 : I(P) \neq I(Q)$

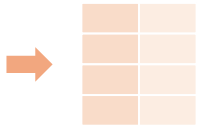
Two populations

Population P
sample size n_P



MI estimate
$\hat{I}(P)$

Population Q
sample size n_Q



MI estimate
$\hat{I}(Q)$

Variables: two categorical variables, X and Y.

Statistical task

Summarise a sample from each population in a count table

Estimate MI from each table

Assess whether sampling noise can explain the difference

Does MI differ between the two populations?

Table icons are schematic. Thesis Sections 1.1 and 2.1; Bossomaier et al. (2016)

Literature review: foundations and the MI-specific step

Existing results provide MI estimates, a Wald test using a standard normal distribution, and moment matching.
This project derives the MI variance sensitivity needed for its degrees of freedom.

	MI estimation and uncertainty	*Wald comparison*	*Welch–Satterthwaite moment matching*
Idea	Estimate MI and first-order sampling variance	Divide the estimated difference by its standard error	Match variance moments to choose degrees of freedom
Key work	Moddemeijer (1989); Paninski (2003)	Wald (1943); Mora and Ruiz-Castillo (2009)	Satterthwaite (1946); Welch (1947); Hutcheson (1970); Pan and Wall (2002)
Required step	Track variation in the variance estimate itself	Assess the fixed cutoff from the standard normal distribution in finite samples	Calculate the MI-specific variance sensitivity

Contribution: Derive that sensitivity and evaluate the resulting degrees of freedom.

Thesis Sections 2.2–2.5 and 3.2–3.3; full references in B13–B14

Method: one statistic, two reference distributions

Both methods use **the same estimated MI difference and standard error**.
Expanded Welch uses estimated degrees of freedom for a Student t distribution.

Shared statistic

$$T = \frac{\widehat{\Delta}_{\mathrm{BC}}}{\widehat{\mathrm{SE}}}, \qquad \widehat{\mathrm{SE}}^2 = \frac{\widehat{V}(P)}{n_P} + \frac{\widehat{V}(Q)}{n_Q}$$

Δ_{BC}: estimated $I(P) - I(Q)$, after leading bias correction.

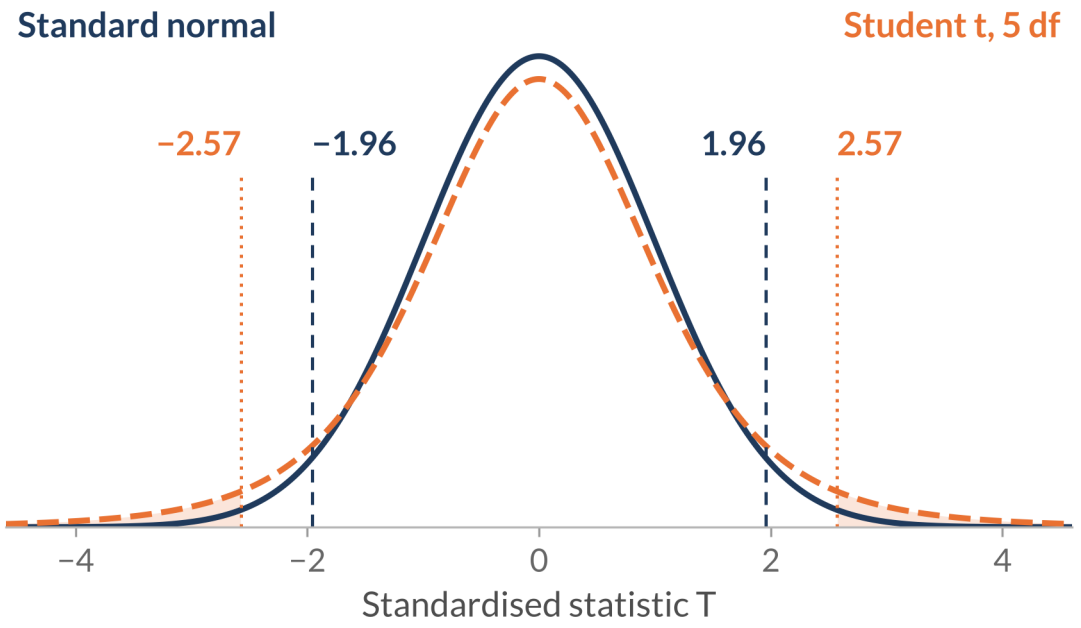
$V(P)$: population variance of pointwise MI;
$\widehat{V}(P)$ estimates it from one sampled table.

Wald test

Standard normal distribution; two-sided 5% cutoff about **±1.96**.

Expanded Welch test

Student t distribution with estimated degrees of freedom; **a larger cutoff**.



Student t distribution with 5 degrees of freedom, for illustration; cutoff about ±2.57.

Approach and contribution

The estimated pointwise-MI variance changes between samples.
We approximate that variation and **use it to choose the degrees of freedom**.

Derivation chain

Cell probabilities change
Move a small amount of probability towards one cell.

→

Margins and pointwise MI change
Track their joint effect.

→

Variance sensitivity, derived here
$g_P(x, y)$ describes the change in $V(P)$.

→

Degrees of freedom
Use the spread of those sensitivities.

Result and approximations

$$\mathrm{Var}\{\widehat{V}(P)\} \approx \frac{\tau^2(P)}{n_P} \quad \Longrightarrow \quad \hat{\nu}_P = \frac{2n_P\widehat{V}(P)^2}{\hat{\tau}^2(P)}$$

$\tau^2(P)$ is the variance of $g_P(X, Y)$. Repeat for Q and combine using Welch–Satterthwaite.

Two approximations evaluated

1. First-order sampling expansion
2. Matching a scaled chi-squared distribution

Thesis Sections 4.1–4.3; exact derivative in B1; g and τ^2 in cell probabilities, with a worked example, on Slides 16–17; moment matching in B2

Evaluation design

Known populations separate approximation accuracy, false positives, and power.
Both tests receive **the same sampled table pairs**.

Research questions

1. **RQ1: Approximation accuracy**
   Sampling moments of $\hat{V}$ and component df, including near independence
2. **RQ2: False positives**
   Equal MI at 0.005, 0.02, and 0.30 nats
3. **RQ3: Power**
   $I(P)$ = 0.02; MI-gap magnitudes 0.005, 0.01, and 0.02 nats

Shared simulation grid

- Four sizes: 2×2, 4×4, 8×8, 16×16
- Uniform, same-skew, and different-skew margins
- Ten sample sizes: 10 to 50,000 observations per population
- **360 settings per study** (4 sizes × 3 margins × 10 sample sizes × 3 MI levels or gaps)
- 100,000 independent table pairs per setting

Model: a copying construction that fixes the margins and solves for the target MI.

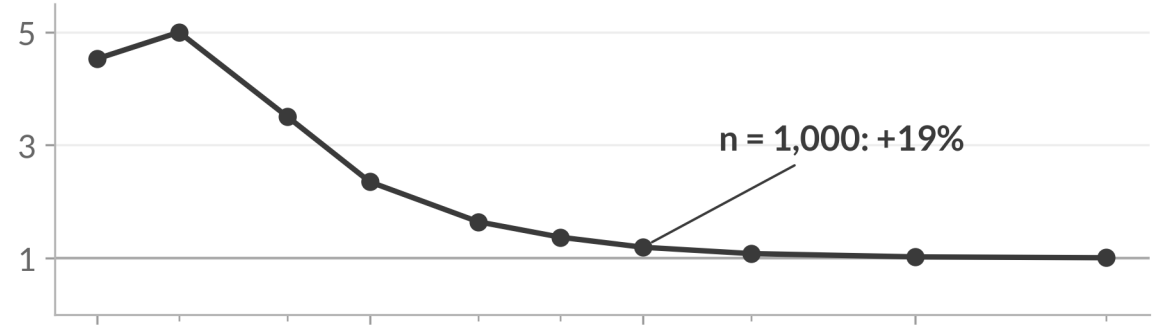
Reporting: usual 0.05 p-value thresholds. Unsuccessful calculations count as non-rejections; calculation success is reported separately.

Morris et al. (2019). Thesis Chapter 5, Tables 5.1–5.3; exact vectors, sample sizes, and requirements in B3

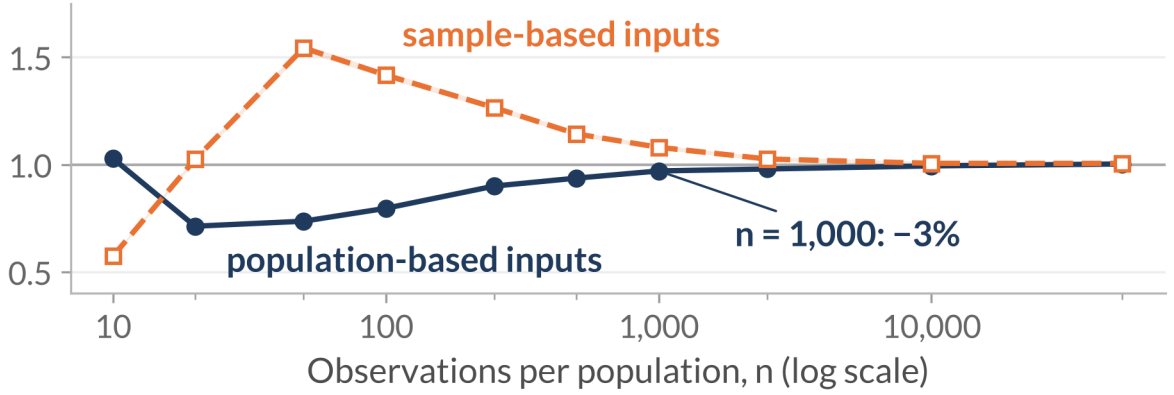
RQ1 results: approximation accuracy

The mean of $\hat{V}$ can **remain inflated** even when its population first-order sampling-variance prediction is accurate.

Mean of $\hat{V}$ / population V



Predicted / observed sampling variance of $\hat{V}$



Different-skew P · 4×4 · population MI 0.02 nats

Mean: rare cells can inflate it.

Newly observed rare cells can have large pointwise-MI deviations, whose squares raise $\hat{V}$.

Sampling variance: linearisation can miss large changes.

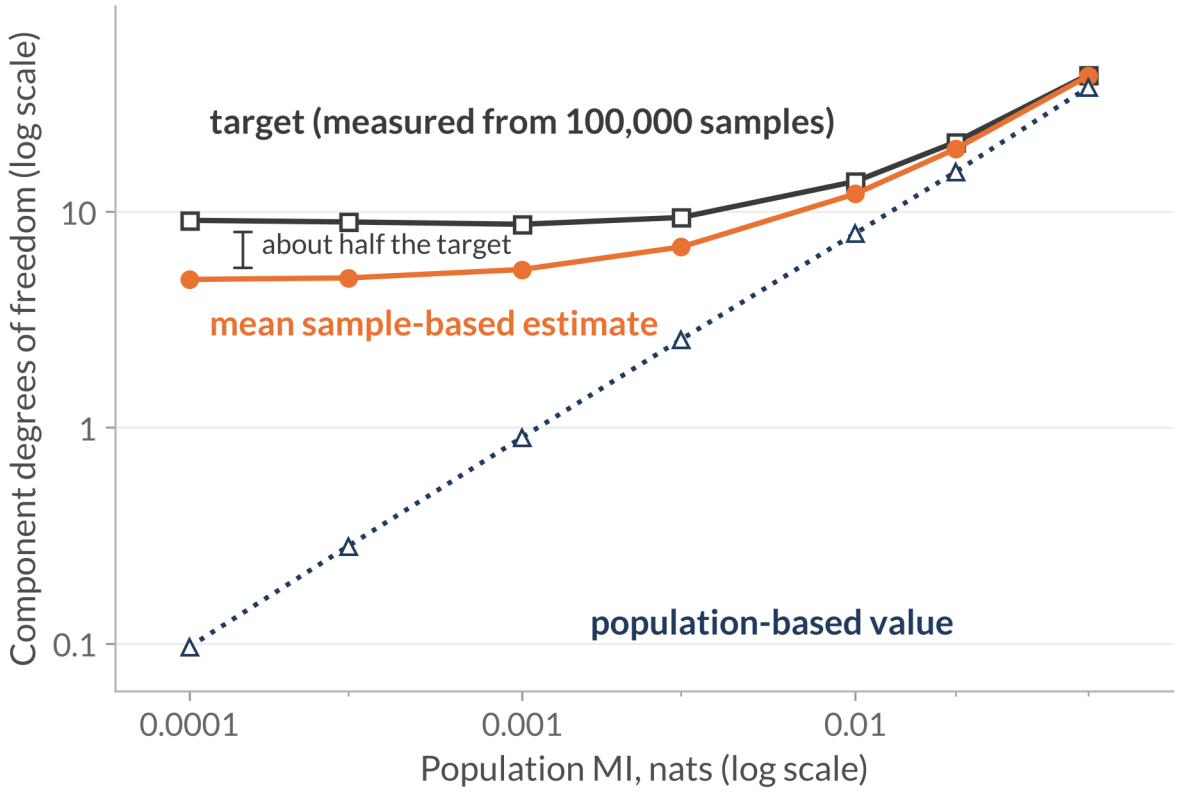
Rare cells appear in some samples and not others; sample-based sensitivities can also overpredict the resulting variation.

At n = 1,000: mean **19% high**;
variance prediction **3% low** with population-based inputs and **8% high** with sample-based inputs.
Both moments enter the degrees of freedom.

Thesis Sections 6.1.1–6.1.2, Figures 6.1–6.2. Bands: approximate 95% Monte Carlo intervals. Other sizes and Q: B4

RQ1 results: near independence

Near independence, first-order theory gives almost no degrees of freedom, while **sampling noise keeps the mean sample-based estimate at about half the target**.



Target: component degrees of freedom from the actual mean and variance of V across 100,000 samples.

Different-skew P · 4×4 · n = 1,000 fixed

Population-based: falls to about n × MI

Inserts known population inputs into the first-order formula. Near independence it is about n × population MI, which heads to zero.

Sample-based: noise keeps it near 5

Quadratic sampling errors create apparent association, which remains in each sample's calculation.

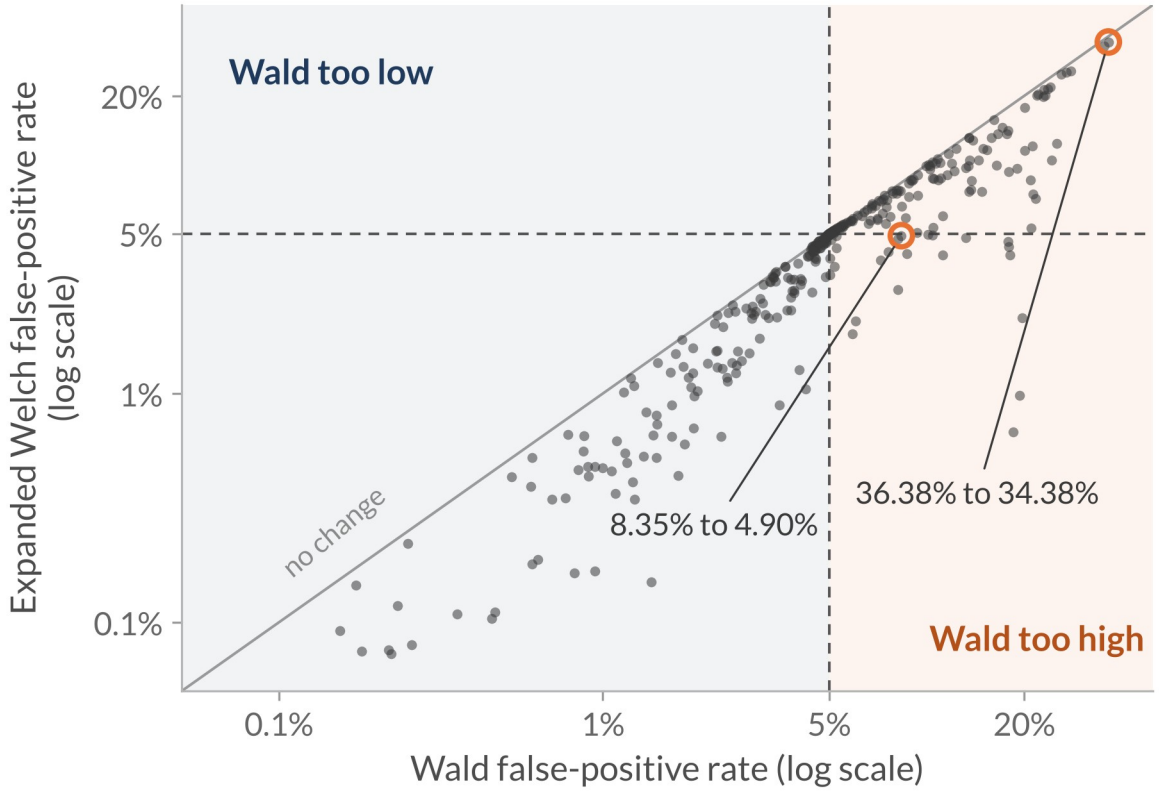
At MI 0.0001: target **≈9.1**;
mean sample-based estimate **≈4.9**;
population-based value **≈0.1**

Thesis Section 6.1.3, Figure 6.3; Appendix A.2. Lines: target $\nu_{P,\mathrm{mom}}$, mean of $\hat{\nu}_P$, population-based $\nu_{P,\mathrm{FO}}$. Bands: approximate 95% Monte Carlo intervals. Local calculation: B5

RQ2 results: false positives across all null settings

A larger cutoff **helps only where Wald rejects too often**;
where Wald already rejects too rarely, it never brings the rate closer to 5%.

All 360 null settings · equal MI in P and Q · each point is one setting



Below the diagonal: Expanded Welch rejects less often. Dashed lines: the 5% target.

Wald too high: 178 settings

Expanded Welch moves closer to 5% in 165, for example 8.35% to 4.90% (uniform 4×4, n = 20, MI 0.02).

Wald too low: 180 settings

Further below 5% in 176 settings; unchanged in four.

Descriptive 4–6% band: 130 versus 136 settings.
Bias-driven excess persists: 36.38% to 34.38% (different-skew 16×16, n = 1,000, MI 0.005).

Tallies of the thesis null study (Sections 6.2.1–6.2.3, Figures 6.4–6.6); two settings sit exactly at 5%. Full grids in B6–B8

RQ2 results: why the cutoff helps, hurts, or cannot fix it

Three kinds of estimation error explain the three patterns on Slide 9.

Three illustrative settings (thesis Table 6.1) · equal MI 0.02 nats · every calculation succeeds

Setting	Bias / SD	Variance ratio	Wald	Expanded Welch
Uniform 4×4, n = 20	≈0	**0.83**	8.35%	4.90%
Same skew 16×16, n = 1,000	≈0	**1.93**	0.60%	0.53%
Different skew 16×16, n = 1,000	**2.06**	1.94	26.01%	24.84%

Bias / SD: shift of the estimated MI difference from zero, in sampling standard deviations. Variance ratio: average squared estimated standard error / observed variance.

Standard error too small

Wald rejects too often. The larger cutoff offsets it: 8.35% to 4.90%.

Standard error too big

Wald already rejects too rarely. The larger cutoff pushes it further below 5%.

Bias in the MI difference

The estimate sits about two SDs from zero. The larger cutoff barely helps: 26.01% to 24.84%.

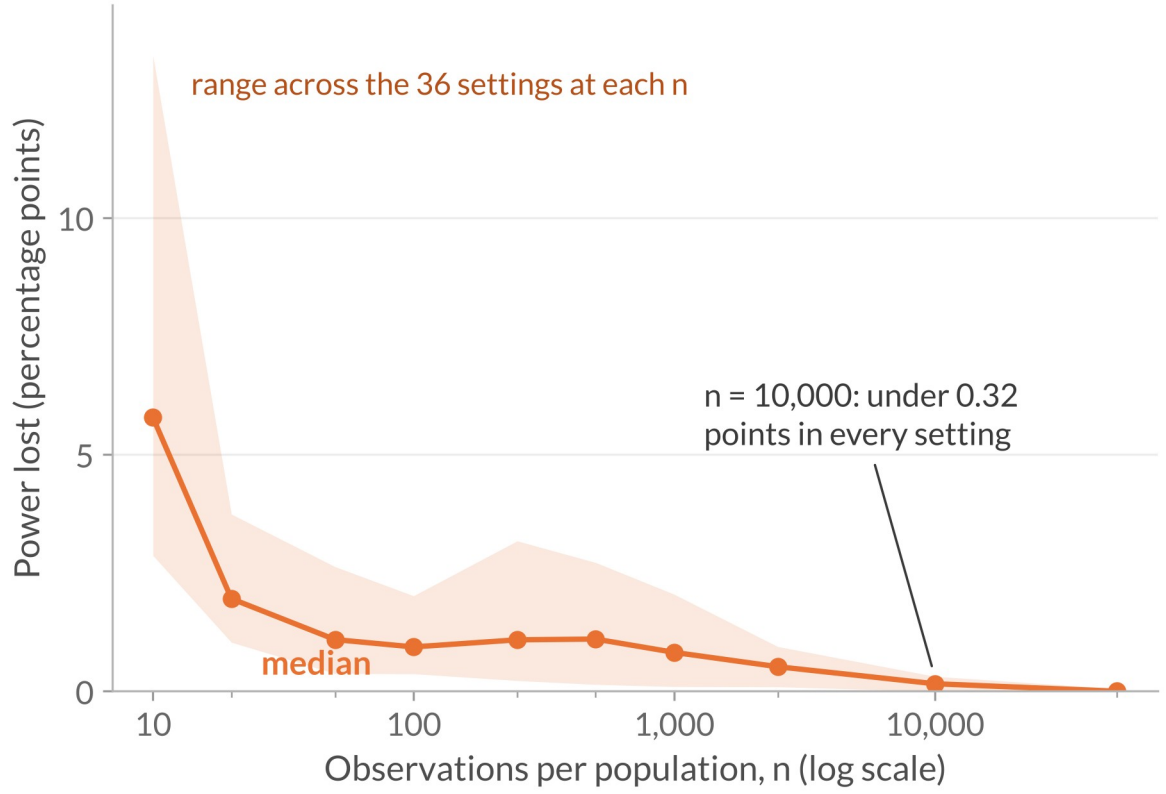
Conclusion: A larger cutoff can offset a standard error that is too small, but it cannot fix bias in the estimated MI difference.

Thesis Section 6.2.4, Table 6.1. Full grids: B6–B8; ordinary Welch comparison: B9

RQ3 results: the power cost of the larger cutoff

Expanded Welch **never has more power than Wald**;
the loss is largest in small samples and under 0.32 points by n = 10,000.

All 360 power settings · MI in P = 0.02 nats · gaps of 0.005, 0.01, and 0.02 nats



Power lost: Wald power minus Expanded Welch power, in percentage points.

The loss is concentrated in small samples

Median loss: about 6 points at n = 10, about 1 point from n = 50 to 1,000, and 0.16 at n = 10,000. Failed calculations add to it at the smallest samples.

Small MI gaps can stay hard to detect

Even at n = 50,000, the 0.005 gap reaches only 79.87%–96.47% power; gaps of 0.01 and 0.02 exceed 99.9%.

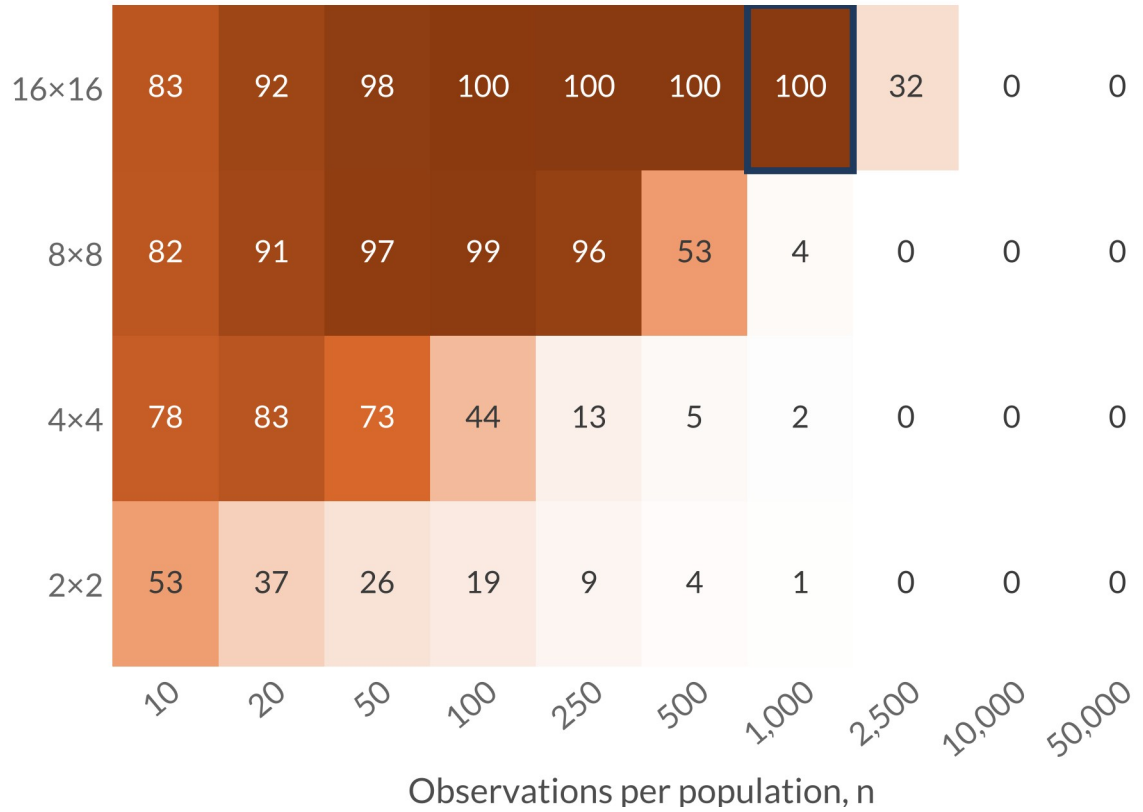
Power: rejecting equal MI when population MI differs, in either direction, at the usual 0.05 threshold.

Tallies of the thesis power study (Sections 6.3.1–6.3.3, Figures 6.7–6.9); unsuccessful calculations count as non-rejections. Full grids in B10–B12

RQ3 results: bias can reverse the estimated direction

Under different-skew margins, sparse tables give rejections that **point to the wrong population**.

Different-skew margins · MI gap 0.01 nats · % of Expanded Welch rejections with the wrong sign



Percentages are rounded. A value of 50% means half the rejections have the wrong estimated sign.

Outlined cell: 16×16, n = 1,000

True difference −0.0100 nats; mean estimate +0.0207. Expanded Welch rejects in 9.66% of repetitions, and over 99.9% of those point the wrong way.

Bias outweighs the gap

A larger cutoff cannot remove bias; more data can. By n = 10,000, under 0.4% of rejections point the wrong way.

All 360 power settings: at least 90% wrong in 28, all with different-skew margins.

Tallies of the thesis power-study diagnostics (Section 6.3.3, Figure 6.9); other gaps show the same pattern. Outlined cell: 9,659 of 9,662 rejections positive

Conclusions and next steps

MI-specific degrees of freedom improve some false-positive rates, while remaining failures identify **estimation accuracy as a priority**.

Contributions

1 Derivation
How the estimated pointwise-MI variance varies between samples

2 Test construction
The Wald statistic with estimated Student t degrees of freedom

3 Empirical evaluation
Accuracy, false positives, power, and calculation success

Findings by research question

RQ1 Approximation accuracy

Approximations improve as samples grow; near independence, first-order theory misses sampling noise.

RQ2 False positives

Helped only where Wald rejected too often; pushed conservative settings further below 5%.

RQ3 Power and direction

At the same nominal 5% threshold, power is no higher; bias can make rejections point the wrong way.

Scope: The evaluation covers one population construction and equal sample sizes, with the Wald test as the main benchmark.

Next steps

- Reduce bias in the estimated MI difference and improve its standard error and degrees of freedom, including near independence.
- Test other association constructions and unequal sample sizes.

Thank you.

Appendix

Variance sensitivity

Perturbation towards one cell: $P_\varepsilon = (1-\varepsilon)P + \varepsilon\delta_{(x,y)}$

Changing probability weights

$$g_P(x,y) = \{\ell_P(x,y) - I(P)\}^2 - V(P)$$

Changing pointwise-MI values and margins

$$+ 2[\ell_P(x,y) - \mathbb{E}_P\{\ell_P(X,Y) \mid X = x\}$$

$$- \mathbb{E}_P\{\ell_P(X,Y) \mid Y = y\} + I(P)]$$

Pointwise MI: $\ell_P(i,j) = \log\{p_{ij}/(p_{i+}p_{+j})\}$

Sensitivity variance: $\tau^2(P) = \mathrm{Var}_P\{g_P(X,Y)\}$

Exact for positive population tables; its use for sampling errors is first order. Thesis Chapter 4

Variance sensitivity in cell probabilities

Each quantity as an expectation, variance, or derivative, and as a sum over the cells of one table

Quantity	Definition	In cell probabilities
Pointwise MI	$\ell_P(x, y) = \log \dfrac{\Pr(X = x,\ Y = y)}{\Pr(X = x)\Pr(Y = y)}$	$\ell_{ij} = \log \dfrac{p_{ij}}{p_{i+}\, p_{+j}}$
MI	$I(P) = \mathbb{E}_P\{\ell_P(X, Y)\}$	$I = \sum_{i,j} p_{ij}\, \ell_{ij}$
Variance of pointwise MI	$V(P) = \mathrm{Var}_P\{\ell_P(X, Y)\}$	$V = \sum_{i,j} p_{ij}(\ell_{ij} - I)^2$
Row mean of pointwise MI	$\mathbb{E}_P\{\ell_P(X, Y) \mid X = i\}$	$\bar{\ell}_{i+} = \dfrac{1}{p_{i+}} \sum_j p_{ij}\, \ell_{ij}$
Column mean of pointwise MI	$\mathbb{E}_P\{\ell_P(X, Y) \mid Y = j\}$	$\bar{\ell}_{+j} = \dfrac{1}{p_{+j}} \sum_i p_{ij}\, \ell_{ij}$
Sensitivity of V to cell (i, j)	$g_P(i,j) = \dfrac{\mathrm{d}}{\mathrm{d}\varepsilon} V(P_\varepsilon) \ \text{ at } \ \varepsilon = 0$ $P_\varepsilon = (1 - \varepsilon)P + \varepsilon\delta_{(i,j)}$	$g_{ij} = (\ell_{ij} - I)^2 - V + 2(\ell_{ij} - \bar{\ell}_{i+} - \bar{\ell}_{+j} + I)$
Variance of the sensitivities	$\tau^2(P) = \mathrm{Var}_P\{g_P(X, Y)\}$	$\tau^2 = \sum_{i,j} p_{ij}\, g_{ij}^2$
Component degrees of freedom	$\nu_P = \dfrac{2[\mathbb{E}\{\widehat{V}(P)\}]^2}{\mathrm{Var}\{\widehat{V}(P)\}} \approx \dfrac{2n_P V(P)^2}{\tau^2(P)}$	$\hat{\nu}_P = \dfrac{2n_P \widehat{V}^2}{\hat{\tau}^2}$

The sensitivities have probability-weighted mean zero. Estimates replace each p_{ij} with n_{ij}/n_P over occupied cells. Worked example: next slide.

Thesis Equations 2.3, 2.4, 4.7, 4.9–4.11, 4.29, 4.36 and 4.41–4.43

Variance sensitivity: a worked example

Thesis table N^P: one cell step by step, then all nine cells

Counts

35	10	5	50
10	20	5	35
5	5	10	20
50	35	20	105

Row and column totals in grey; $\hat{p}_{ij} = n_{ij} / 105$.

Sensitivities of all nine cells

0.422	−1.048	−1.382
−1.048	0.758	−1.134
−1.382	−1.134	1.620

Calculated from thesis Equation 4.41. Their weighted mean is zero.

Cell (1, 1), step by step

1 Pointwise MI

$$\hat{\ell}_{11} = \log \frac{35/105}{(50/105)(50/105)} = 0.385$$

2 MI and variance of pointwise MI

$$\hat{I} = \sum_{i,j} \hat{p}_{ij} \hat{\ell}_{ij} = 0.137, \qquad \hat{V} = \sum_{i,j} \hat{p}_{ij} (\hat{\ell}_{ij} - \hat{I})^2 = 0.272$$

3 Row 1 mean (column 1 is the same, by symmetry)

$$\bar{\ell}_{1+} = \frac{35(0.385) + 10(-0.511) + 5(-0.644)}{50} = 0.103$$

4 Sensitivity

$$\hat{g}_{11} = (0.385 - 0.137)^2 - 0.272 + 2(0.385 - 0.103 - 0.103 + 0.137) = 0.422$$

5 All nine cells

$$\hat{\tau}^2 = \sum_{i,j} \hat{p}_{ij} \hat{g}_{ij}^2 = 0.932, \qquad \hat{\nu}_P = \frac{2(105)(0.272)^2}{0.932} = 16.65$$

Thesis Section 4.4.1 (Equation 4.47). Cell (1, 1) adds $(35/105)(0.422)^2 = 0.059$ to the sum in step 5. Results use unrounded values

Moment matching and combined degrees of freedom

Component degrees of freedom

$$\nu_{P,\,\mathrm{mom}} = \frac{2[\mathbb{E}\{\widehat{V}(P)\}]^2}{\mathrm{Var}\{\widehat{V}(P)\}}, \qquad \widehat{\nu}_P = \frac{2n_P\widehat{V}(P)^2}{\widehat{\tau}^2(P)}$$

Combined degrees of freedom

$$\widehat{\nu} = \frac{\{\widehat{V}(P)/n_P + \widehat{V}(Q)/n_Q\}^2}{\{\widehat{V}(P)/n_P\}^2/\widehat{\nu}_P + \{\widehat{V}(Q)/n_Q\}^2/\widehat{\nu}_Q}$$

Moment matching does not establish the standard normal numerator, or its independence from the variance estimates, that an exact Student t distribution requires.

Satterthwaite (1946). Thesis Sections 4.1 and 4.3

Population construction, grid, and calculation requirements

$$p_{ij}(\lambda) = (1-\lambda)a_i a_j + \lambda a_i \mathbf{1}\{i=j\}, \qquad 0 \le \lambda < 1$$

Parameter grid

- k = 2, 4, 8, 16; uniform, same-skew, or different-skew margins
- Same skew: both use $m_k(0.8)$. Different skew: P uses $m_k(0.7)$, Q uses $m_k(0.8)$, where $m_k(\rho) = (\rho, (1-\rho)/(k-1), \ldots)$
- Null: shared MI 0.005, 0.02, 0.30. Power: $I(P)$ = 0.02; $I(Q)$ = 0.025, 0.03, 0.04

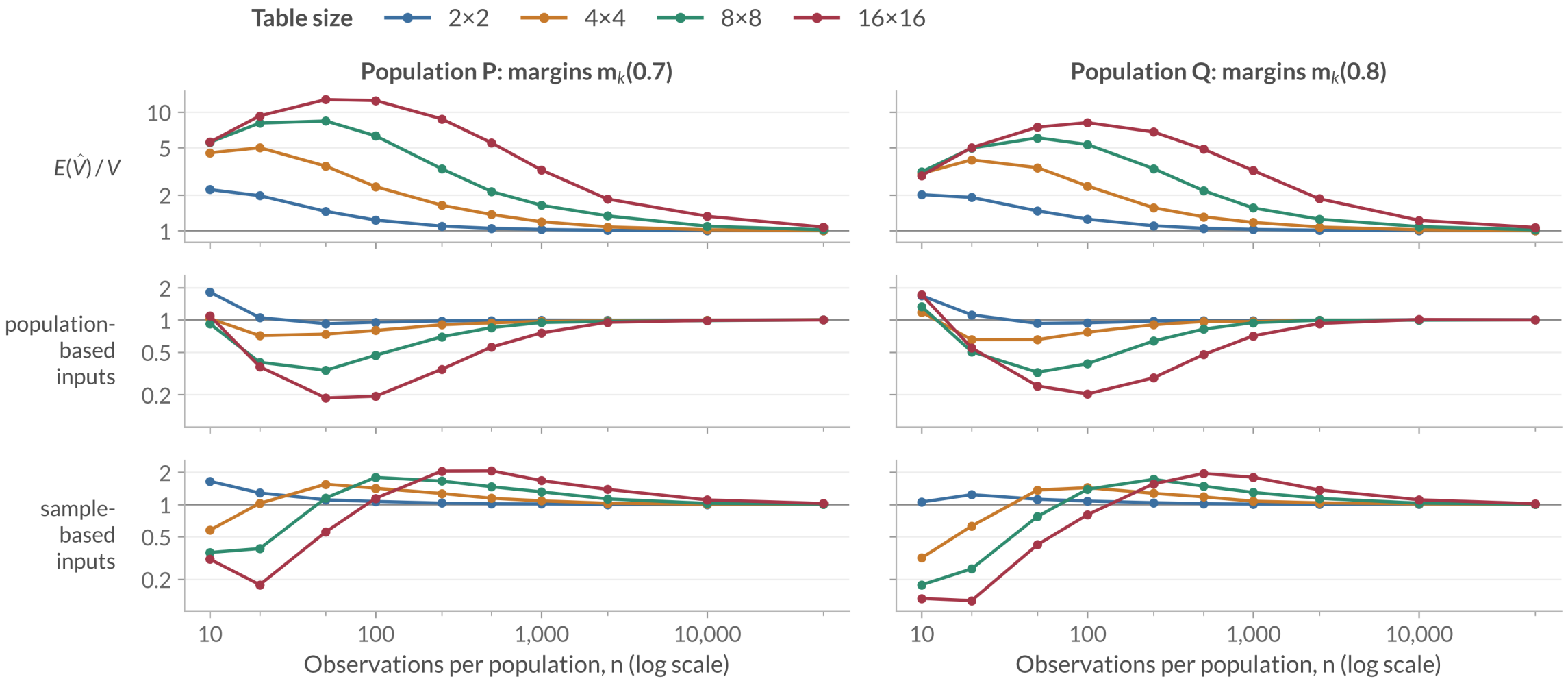
Calculation requirements and rates

- Each null or power study has 360 settings, 100,000 independent pairs per setting
- A Wald calculation needs a finite difference and a positive finite standard error. Expanded Welch also needs positive finite degrees of freedom.
- Rate = rejections from successful calculations / all repetitions. Calculation success is reported separately.

n = 10, 20, 50, 100, 250, 500, 1,000, 2,500, 10,000, 50,000 per group

Approximation ratios across table sizes

Different-skew margins, shared MI 0.02 nats; a ratio of one indicates agreement



Rows: mean of $\hat{V}/V$; population-based and sample-based variance predictions / observed variance. Thesis Figure 6.1

Near-independence noise calculation

Setting: 4×4, n = 1,000, MI 0.0001; $d = 9$ and $\kappa = 2nI(P) \approx 0.2$

$$G = 2n\hat{I}(P) \;\dot{\sim}\; \chi^2_d(\kappa), \qquad \hat{\nu}_P \approx G/2$$

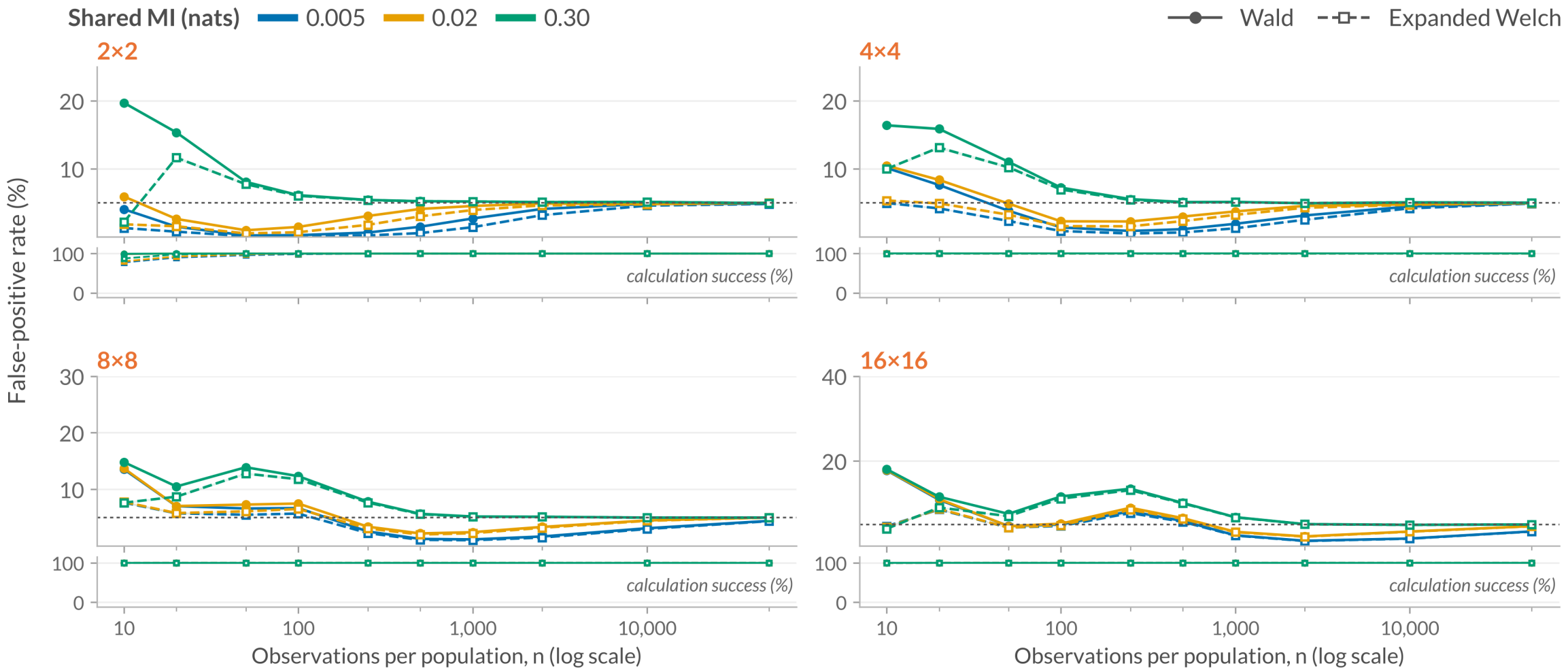
In $G, \hat{I}(P)$ is the uncorrected plug-in MI, not a bias-corrected estimate.

Quantity	Local prediction	Observed (P)
Sample-based estimate (mean component df)	4.60	4.86
Target (from sampling moments)	9.00	9.12
Mean sample-based variance prediction / observed sampling variance	1.96	1.97

Local approximations under fixed dimensions and positive limiting cell probabilities. Thesis Appendix A.2, Section 6.1.3

Full null results: uniform margins

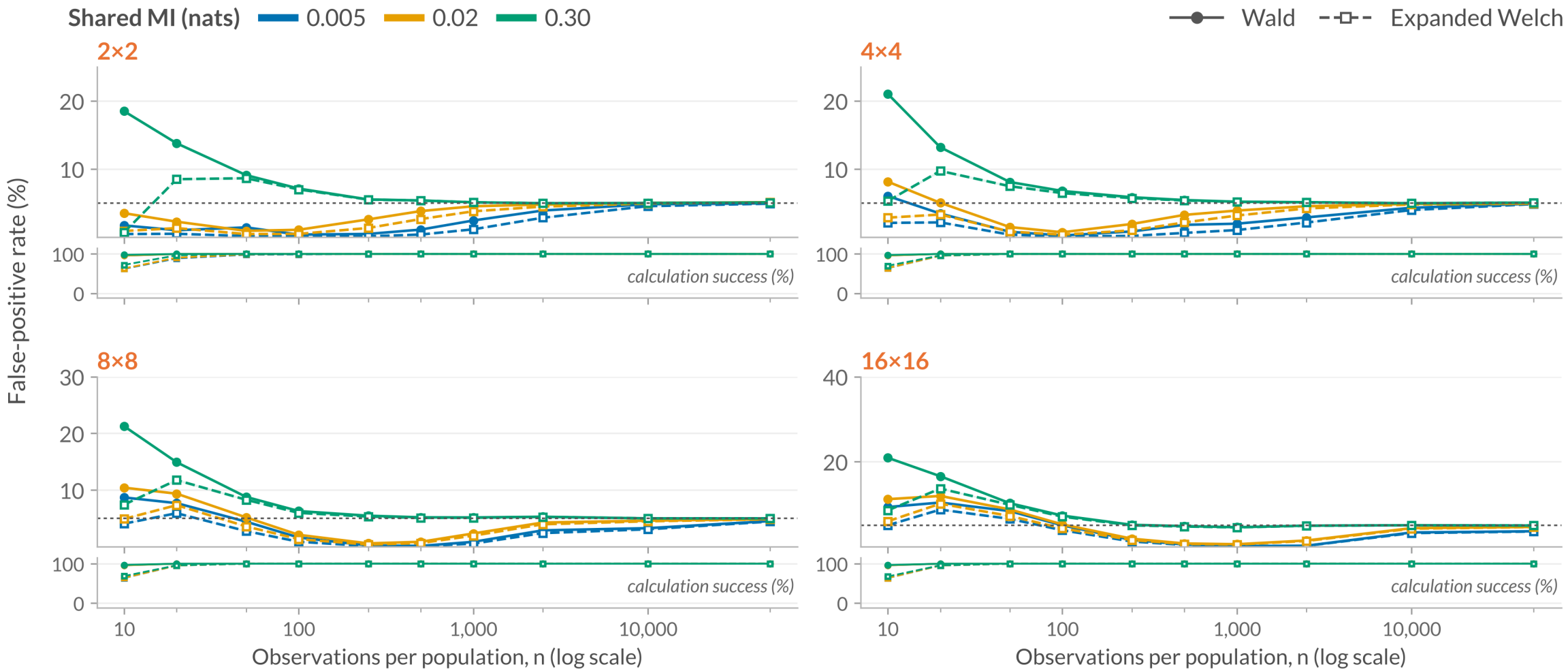
Equal MI in P and Q; uniform margins in both populations



Rates divide rejections by all 100,000 repetitions; unsuccessful calculations count as non-rejections. Requirements in B3. Thesis Figure 6.4

Full null results: same-skew margins

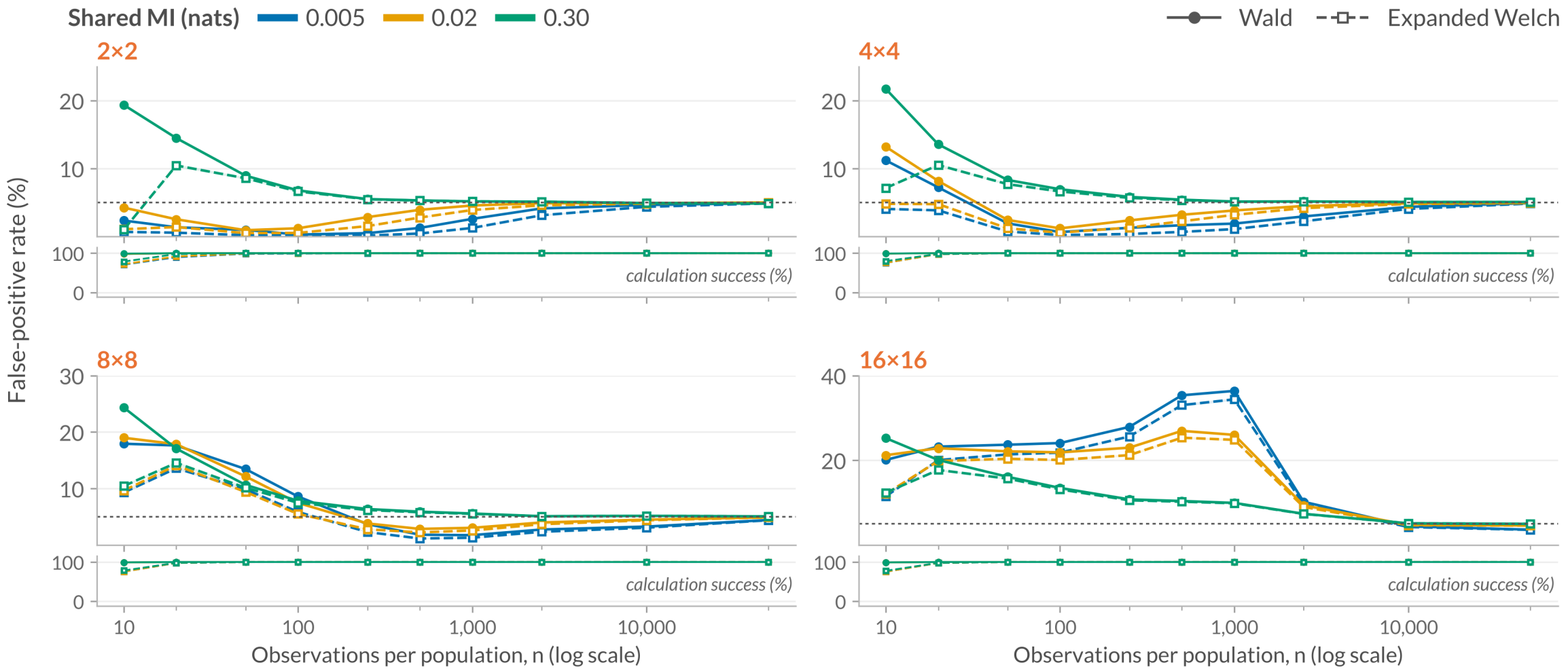
Equal MI; same-skew margins $m_k(0.8)$ in P and Q



Rates divide rejections by all 100,000 repetitions; unsuccessful calculations count as non-rejections. Requirements in B3. Thesis Figure 6.5

Full null results: different-skew margins

Equal MI; $m_k(0.7)$ in P and $m_k(0.8)$ in Q



Rates divide rejections by all 100,000 repetitions; unsuccessful calculations count as non-rejections. Requirements in B3. Thesis Figure 6.6

Ordinary Welch degrees of freedom

Same statistic and sampled pairs; ordinary Welch gives each component $n - 1$ degrees of freedom; shared MI 0.02 nats

Setting	Wald	Ordinary Welch	Expanded Welch
Uniform 4×4, n = 20	8.35%	7.35%	4.90%
Uniform 4×4, n = 50	4.87%	4.57%	3.24%
Different skew 16×16, n = 1,000	26.01%	25.95%	24.84%

False-positive rates; nominal target 5%.

n = 20

Ordinary Welch reduces the excess less than Expanded Welch.

n = 50

Wald is already close to 5%; Expanded Welch is the most conservative.

Different skew 16×16

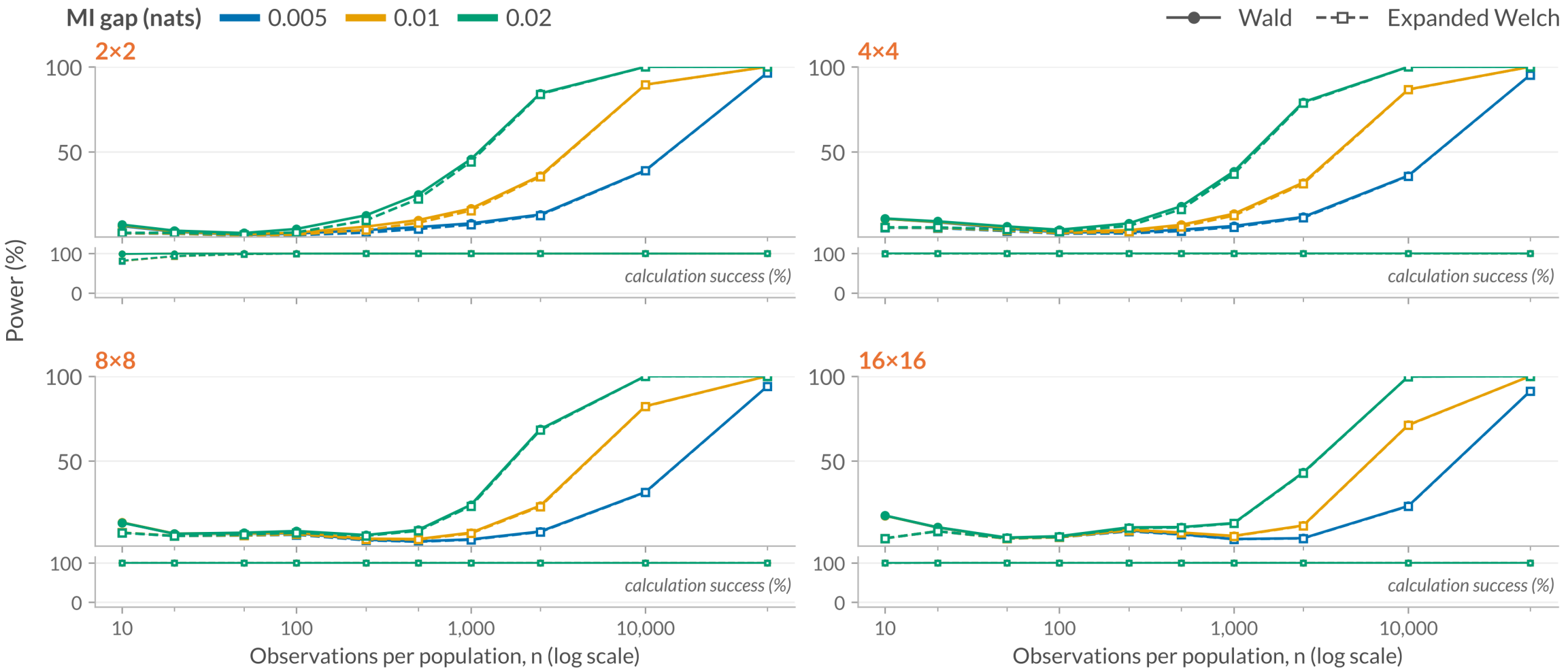
Neither choice removes the excess caused by bias.

Conclusion: MI-specific degrees of freedom change the results, but a larger reduction in rejection does not consistently improve calibration.

Same 100,000 sampled pairs per setting as the null study; every calculation succeeds. Thesis Section 6.2.5, Table 6.2

Full power results: uniform margins

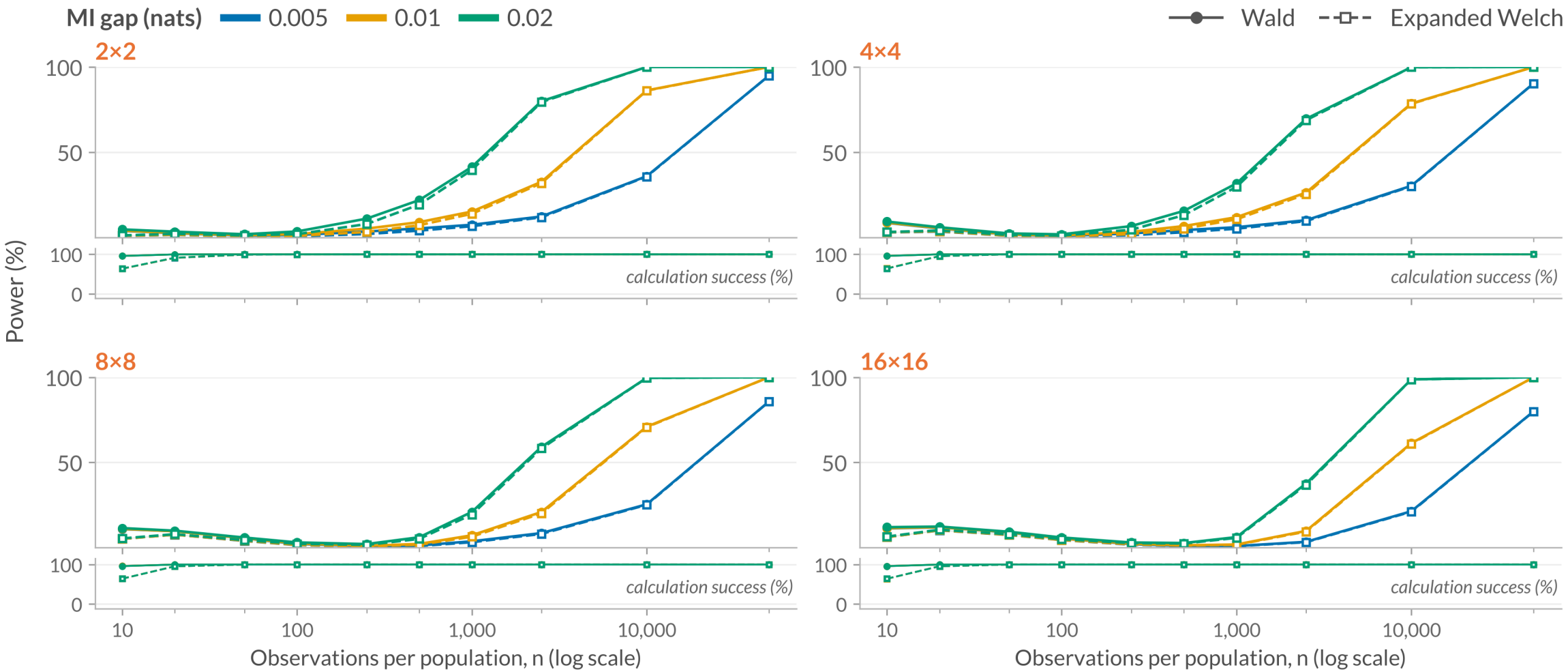
Uniform margins; MI in P = 0.02 nats



Rates divide rejections by all 100,000 repetitions; unsuccessful calculations count as non-rejections. Requirements in B3. Thesis Figure 6.7

Full power results: same-skew margins

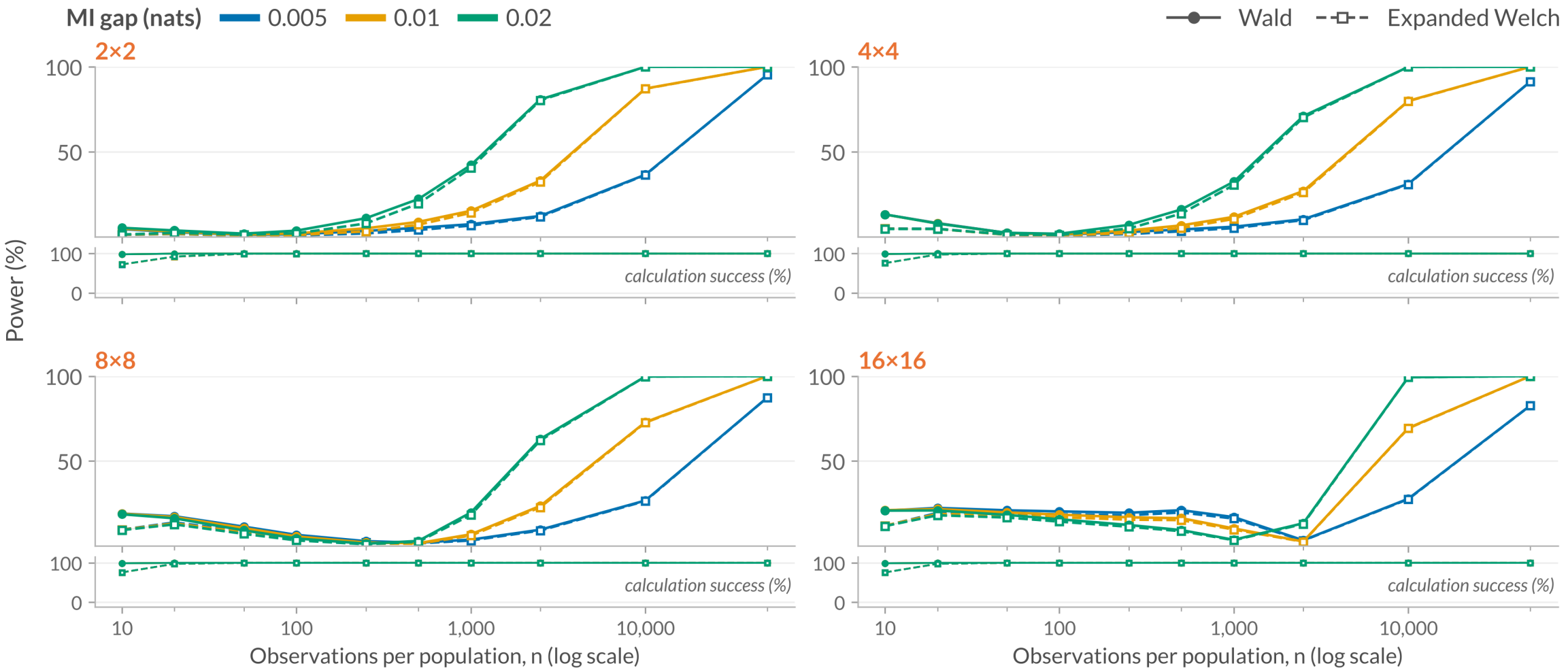
Same-skew margins $m_k(0.8)$ in P and Q; MI in P = 0.02 nats



Rates divide rejections by all 100,000 repetitions; unsuccessful calculations count as non-rejections. Requirements in B3. Thesis Figure 6.8

Full power results: different-skew margins

$m_k(0.7)$ in P, $m_k(0.8)$ in Q; MI in P = 0.02 nats



Rates divide rejections by all 100,000 repetitions; unsuccessful calculations count as non-rejections. Requirements in B3. Thesis Figure 6.9

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